Agenda

- Last time
  - Indicator random variables
  - Standardization of data

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Standardization in one dimension

Raw deviation	→	Standardized deviation
$X - \mu$		$Z = (X - \mu) / \sigma$
unit-dependent		unit-free: “2 standard deviations above mean”

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Agenda

- Last time
  - Indicator random variables
  - Standardization of data
  - Correlation coefficient
  - Correlation is not causation
  - “Bi-Linearity” of covariance
- Today
  - Inequalities
  - Weak Law of “large numbers”

Cauchy–Schwarz inequality

For vectors, $|\langle x,y\rangle|\leq\|x\|\|y\|$.

Inner product cannot be arbitrarily large relative to the lengths of the vectors

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$E[(X-\mu_X)(Y-\mu_Y)$

$var(X) = E[X^2]-(E[X])^2$

A formula similar to that of the mean: Cov(X,Y)=E[XY]−E[X]E[Y]

$E[(X-\mu_X)(Y-\mu_Y)]$

$= E[XY - \mu_X Y - \mu_Y X + \mu_X\mu_Y]$

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$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$

Covariance and variance

- Symmetric: Cov(X,Y) = Cov (Y,X)
- “Bilinear”: Linear in each argument separately
  - Cov(aX,Y) = a Cov(x,y)
    - Recall: Var(aX) = a*a Var(X)
  - Cov(X+Z, Y) = Cov(X,Y) + Cov(Z,Y)
- Cov(X,X) = Var(X)
- Var(X+Y) =Var(X)+Var(Y)+2Cov(X,Y) ← Must know

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Independence ⟹ zero covariance

- A formula similar to that of the mean: Cov(X,Y)=E[XY]−E[X]E[Y]

$$E[XY] = \sum_i \sum_j x_i\, y_j P\{X = x_i,\ Y = y_j\} = \sum_i \sum_j x_i\, y_j P\{X = x_i\}P\{Y = y_j\}$$
$$= \sum_i x_i\, P\{X = x_i\} \sum_j y_j\, P\{Y = y_j\} = E[X]E[Y]$$

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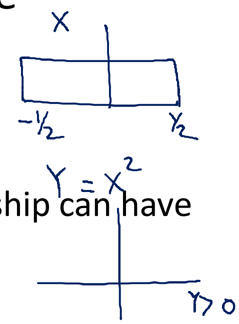
Zero covariance ⇏ Independence



- Covariance only measures **linear** association.
  - Try $Y = X^2$ with X symmetric about 0
- A perfectly deterministic but non-monotonic relationship can have zero covariance.

$$\mathrm{Cov}(X, Y)$$
$$= \mathrm{Cov}(X, X^2)$$
$$= E[X \cdot X^2] - E[X] \cdot E[Y]$$
$$\underbrace{E[X^3]}_{0} \qquad 0$$

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Probabilistic inequalities

- Sometimes we know only the mean or variance of a random variable, and want the probability that it takes on a certain value.
  - The exact probability cannot be computed without assumptions
- But we can obtain upper or lower bounds on that probability, and those bounds are often enough to make a decision

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Markov's inequality

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3
- What is the probability that a randomly chosen student gets an offer of 110×10^3, or more?
- Let *X* be a random variable that takes only **non-negative** values. For any *a* > 0:

$$P\{X \geq a\} \leq E[X]/a$$

Markov's Inequality

$$E[X] = \int_0^\infty x\, f_X(x)\, dx = \int_0^a x\, f_X(x)\, dx + \int_a^\infty x\, f_X(x)\, dx$$

Similar proof for discrete case also

$$\geq \int_a^\infty x\, f_X(x)\, dx$$

$$\geq \int_a^\infty a\, f_X(x)\, dx$$

$$= a \int_a^\infty f_X\, (x)\, dx = a\, P\{X \geq a\}$$

$$\therefore\ P\{X \geq a\} \leq E[X]/a$$

Markov's inequality

$P\{X \geq a\} \leq E[X]/a$

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3
  - What is the probability that a randomly chosen student gets an offer of 110×10^3, or more?
- If we assume that the first statement means
  - Choose a random student
  - Let X = salary offered to the student. E[X] = 100×10^3
  - Then the probability a randomly chosen student gets 110×10^3 is equal to or less than 90.9%

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Chebyshev's inequality

- Suppose the expected value of the salary offered to a CSE BTech-4 student at IITB is 100×10^3 and the variance is 50×10^3
  - What is the probability that a student's salary lies between 90×10^3 and 110×10^3?
- For a random variable X with mean μ and variance σ², and any k > 0

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

$$P\{\left|X - 100 \times 10^3\right| \geq 10 \times 10^3\} \leq \frac{50 \times 10^3}{10 \times 10 \times 10^6} = 0.0005 \approx 0.05\%$$

$$\therefore\ P\{\left|X - 100 \times 10^3\right| < 10 \times 10^3\} = 1 - 0.05\% = 99.5\%$$

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Chebyshev's Inequality

$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$

- For a random variable X with mean μ and variance σ^2, and any $k > 0$:
- Apply Markov's inequality to $(X-\mu)^2$ since it is non-negative

$$P\{(X-\mu)^2 \geq k^2\} \leq E[(X-\mu)^2]/k^2 = \sigma^2/k^2$$

$$\therefore\ P\{|X-\mu| \geq k\} \leq \sigma^2/k^2$$

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Chebyshev’s Inequality

- For X with mean μ and variance σ^2, and any $k > 0$:

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- Replacing k by $k\sigma$ gives the form that is usually quoted — the bound now reads in units of standard deviations:

$$P\{|X-\mu| \geq k\sigma\} \leq \frac{1}{k^2}$$

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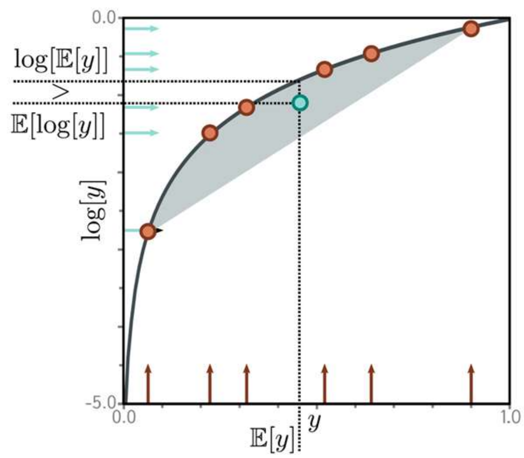
Jensen's inequality

Illustration for log function

- For any concave function
  - $f[E[y]] \geq E[f[y]]$
  - Draw a straight line between any two points, it lies under the curve
- Any weighted sum of the six points must line in the gray region
- Sample mean on the x-axis is shown



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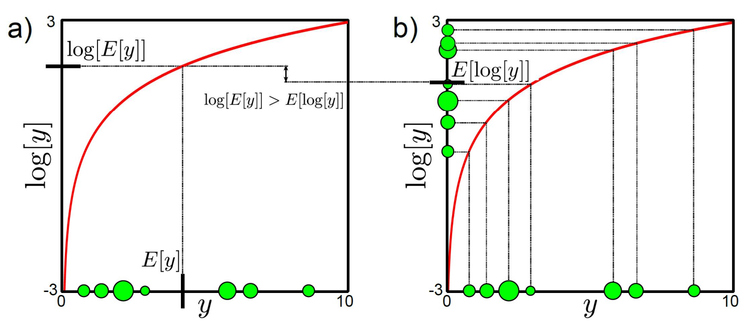
Jensen's Inequality



$$E\left[\log[y]\right] \leq \log\left[E[y]\right]$$ or... $$\sum_y q(y)\log[y] \leq \log\left[\sum_y q(y)y\right]$$

similarly $$\int q(y)\log[y]dy \leq \int \log[q(y)]dy$$

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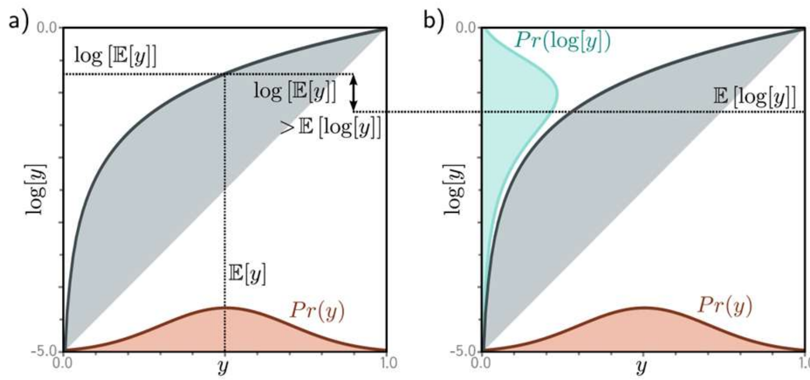
Continuous case



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